# DIABETIC RETINOPATHY (DR) PROGRESSION DIGITAL TWIN SYSTEM ARCHITECTURE

**LAYER 1: DATA INGESTION LAYER**

- **Portable Retinal Fundus Images** (JPG/PNG from field cameras)
- **Longitudinal Electronic Health Records (EHR)** (HbA1c %, Systolic BP, eGFR kidney score, Diabetes Duration, Medication Adherence)
- **Wearable Telemetry** (Continuous Glucose Monitor 5-minute CGM streams, Heart Rate, Steps)

Clinical & Sensor Data Bypass

Image Stream

**LAYER 2: VISION AI & EXPLAINABILITY ENGINE**

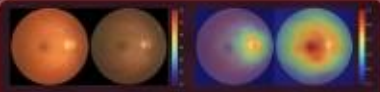
- **Preprocessing Pipeline** (Circular ROI Crop + Ben Graham Illumination Normalization + LAB CLAHE Contrast Enhancement)
- **Pretrained EfficientNet-B3 CNN** (with custom 5-class ICDR classification head, Grade 0: No DR to Grade 4: Proliferative DR)
- **GradCAM Saliency Engine** (Hooks conv_head layer to overlay visual heatmaps showing microaneurysms and exudate lesions)

ICDR Grade & Saliency Maps

**LAYER 3: DIGITAL TWIN SIMULATION ENGINE**

- **8-Factor Evidence-Based Composite Risk Calculator** (Weighted by UKPDS, DCCT, WESDR landmark trials)
- **5-State Markov Chain Monte Carlo (MCMC) Simulator** (1,000 stochastic runs over 5-year horizon, adjusting transition speeds via patient risk multiplier)
- **Evidence-Based Screening Recommender** (Maps risk scores to personalized checkup intervals: 24, 12, 6, 3 months, or Emergency Specialist Referral)

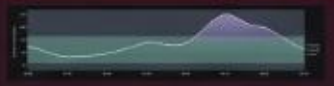
**LAYER 4: PRESENTATION & CLINICAL DASHBOARD** (Streamlit + Plotly)



**Panel A: Eye Comparison & AI Confidence** (Original vs. GradCAM Heatmap + AI Confidence Bar)



**Panel B: Risk Speed-Gauge & Radar Chart** (0-100 Multi-Zone Gauge & 8-Axis Clinical Risk Radar)



**Panel C: Glucose Circadian Curve** (7-Day CGM with 70-180 mg/dL Target Band)



**Panel D: 5-Year Progression & Reports** (5-Yr Progression Probability Area Chart, Urgency Badges, Download .txt & .html